

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 1

Case Study: Modelling Daily Exchange Rate Returns

Application area: *Finance — Foreign Exchange*

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FRED — Federal Reserve Bank of St. Louis (<https://fred.stlouisfed.org/series/DEXBZUS>): series DEXBZUS (Brazilian Reais per U.S. Dollar, daily, 2003–2024). Download via the `fredr` package in R. A free API key is required: https://fred.stlouisfed.org/docs/api/api_key.html. FRED returns NA on holidays — no carry-forward cleaning is needed.

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1. Background and Objectives

The daily USD/BRL exchange rate is one of the most volatile among emerging-market currencies, reflecting Brazil's commodity dependence, political cycles, and external financing conditions. This tutorial applies the Box–Jenkins ARMA framework to log-returns of the USD/BRL rate and evaluates whether simple time series models can capture the dynamics of this series.

Learning objectives: (i) Download and prepare financial time series data from FRED; (ii) handle missing observations (holidays) correctly; (iii) apply stationarity tests and ACF/PACF identification; (iv) estimate and validate ARMA models; (v) produce out-of-sample forecasts and evaluate accuracy.

2. Data Preparation

Data are sourced from FRED using the series DEXBZUS. FRED is preferred over Yahoo Finance for this series because it returns NA on US Federal and Brazilian holidays rather than carrying forward the previous day's price — which would introduce spurious zero-return observations.

```
# install.packages(c("fredr", "xts", "tseries", "forecast", "urca",  
#                  "FinTS", "strucchange", "moments"))  
library(fredr); library(xts); library(tseries); library(forecast)  
library(urca); library(FinTS); library(strucchange); library(moments)  
  
# Set your free FRED API key (register at fred.stlouisfed.org)  
fredr_set_key("YOUR_FRED_API_KEY")  
  
# Download DEXBZUS: Brazilian Reais per U.S. Dollar, daily  
raw <- fredr(series_id      = "DEXBZUS",  
             observation_start = as.Date("2003-01-01"),  
             observation_end   = as.Date("2024-12-31"),  
             frequency        = "d")  
  
# Convert to xts; remove any weekend rows (guard)  
brl <- xts(raw$value, order.by = raw$date)  
brl <- brl[.indexyday(brl) %in% 1:5]  
  
# FRED already returns NA on holidays --- no carry-forward fix needed  
log_brl <- log(brl)  
ret_brl <- diff(log_brl)[-1] # NA propagates automatically on holiday  
                        days  
  
# Complete observations for statistical tests  
rbrl <- as.numeric(na.omit(ret_brl))
```

```
lbrl <- as.numeric(na.omit(log_brl))
```

FRED returns NA on days when the market was closed (US Federal holidays and most Brazilian holidays). Calling `diff()` on the log level propagates NA correctly: if either day t or $t - 1$ is a holiday, the log-return $r_t = \log(P_t/P_{t-1})$ is also NA. This is the correct behaviour — there is no observed return when the market was closed. For all statistical tests, extract complete observations with `na.omit()`.

```
df <- data.frame(
  date      = index(brl),
  price     = as.numeric(brl),
  log_price = as.numeric(log_brl),
  log_return = c(NA, as.numeric(ret_brl))
)
write.csv(df, "brl_usd_fred.csv", row.names = FALSE, na = "")
# na = "" writes holiday rows as empty cells (clean for Excel)
```

3. Exercises

3.1. Descriptive analysis

Plot the log USD/BRL level and the log-returns (note: gaps in the plot correspond to holidays coded as NA). Compute and report for the complete (non-missing) return observations: mean, standard deviation, skewness, and excess kurtosis. What do these statistics tell you about the distribution of USD/BRL returns?

Use `na.omit(ret_brl)` before computing statistics. Excess kurtosis > 0 confirms fat tails — extreme depreciations and appreciations occur far more often than a normal distribution predicts. Major spikes are visible in 2002 (election crisis), 2008 (global financial crisis), 2015–16 (Dilma impeachment), and 2020 (Covid shock). These features motivate GARCH modelling in Tutorial 6.

3.2. Stationarity

- (a) Apply the ADF test (with constant, BIC lag selection) to the log level `lbrl`. Interpret the result.
- (b) Apply the ADF test to the log-returns `rbrl`. What do you conclude about the integration order of the series?
- (c) Apply the KPSS test to both series. Do the conclusions agree with the ADF?

Use `ur.df(lbrl, type="drift", selectlags="BIC")` from the `urca` package. The 5% critical value for the drift specification is -2.86 . Compare the ADF and KPSS conclusions: if both agree the series is $I(1)$ in levels and $I(0)$ in returns, first-differencing (log-returns) is the correct transformation.

3.3. ACF/PACF identification

Plot the ACF and PACF of the log-returns `rbrl` up to lag 20. Based on the plots, suggest two or three candidate $ARMA(p, q)$ specifications. Explain your reasoning. What does near-zero autocorrelation imply about market efficiency?

3.4. Estimation and model selection

- (a) Estimate $ARMA(0,0)$, $AR(1)$, $MA(1)$, and $ARMA(1,1)$ by MLE using `Arma(rbrl, order=c(p,0,q))`.
- (b) Compute AIC and BIC for each. Which model is selected by BIC? Use `auto.arima(rbrl, ic="bic")` to confirm.
- (c) Inspect the characteristic roots of the selected model using `autoplot()`. Are all roots outside the unit circle?

3.5. Validation

For the BIC-selected model:

- (a) Apply the Ljung–Box test to residuals at lags 5, 10, and 20 (use `fitdf=p+q` to correct degrees of freedom). Is there evidence of remaining serial correlation?
- (b) Apply the Ljung–Box test to *squared* residuals at lags 5 and 10, and the ARCH LM test (`ArchTest` from `FinTS`). What does this tell you?
- (c) Plot CUSUM and CUSUMSQ of recursive residuals (`efp` from `strucchange`). Is there evidence of instability in the mean or variance?

The Ljung–Box test on *residuals* should not reject white noise — the ARMA model captures the mean dynamics. The Ljung–Box test on *squared residuals* should strongly reject white noise — evidence of ARCH effects. The CUSUMSQ should also reject variance constancy. These two failures motivate GARCH modelling in Tutorial 6.

3.6. Forecasting

Re-estimate the selected model on the training period 2003–2023 (complete observations only). Produce rolling 1-step-ahead forecasts for all non-missing trading days in 2024. Compute RMSFE, MAE, and Theil’s U relative to a random walk benchmark (forecast = 0 for all days). Does the ARMA model outperform the random walk? Relate your answer to the efficient market hypothesis.

Use only complete (non-NA) return observations for both training and testing — extract them with `na.omit()` before splitting the sample. This ensures no NA values enter the rolling forecast loop and that the RMSFE is computed only over actual trading days.

4. Core References

Box, G.E.P. et al. (2015). *Time Series Analysis*, 5th ed. Wiley. Fama, E.F. (1965). *Journal of Business*, 38, 34–105. Lo, A.W. & MacKinlay, A.C. (1988). *Review of Financial Studies*, 1, 41–66. Morettin, P.A. & Valls Pereira, P.L. (2021). *Econometria Financeira*, 4th ed. Blucher.

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